

A new numerical model of the HF antenna of the Mars Reconnaissance Orbiter's (MRO) Shallow Radar (SHARAD) and first results from a test at a high-angle spacecraft roll

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ABSTRACT

This communication deals with a newly developed electromagnetic model of the Mars Reconnaissance Orbiter Shallow Radar (SHARAD) HF antenna and its interaction with the spacecraft. The model has been developed considering the effect of spacecraft materials and components and the orientation of appendages such as the solar arrays and high-gain antenna. The model appears very useful in designing new operational scenarios of the SHARAD sub-surface sounder, confirming the ability to increase up to 10 dB the signal-to-noise ratio in radar sounding performances by altering the geometry of the observations. We show results from a first test wherein the spacecraft roll angle was set to 120° while obtaining an observation over a ground track previously observed at 0° roll angle, and the expected performance improvement was achieved.

1. Introduction

The SHALLOW RADAR (SHARAD) instrument aboard Mars Reconnaissance Orbiter (MRO) started its science phase around Mars at the end of 2006, and has substantially contributed to the success of the MRO mission by discovering, among other insights, glacial ice within the mid-latitudes, the three-dimensional layered structure of the polar caps, and subsurface details of some volcanic terrain and sedimentary deposits (Seu et al., 2007a; Putzig et al., 2023). SHARAD has also enabled the retrieval of important data on the permittivity and loss of Martian near-surface materials and the Martian ionosphere (Campbell and Morgan, 2018; Bramson et al., 2015; Campbell et al., 2021b).

SHARAD transmits a frequency-modulated “chirp” pulse having an 85 μ s duration, a frequency band of 15 to 25 MHz, and a power of 10 W. Correlation of the received signal with a reference chirp waveform produces an equivalent compressed pulse of 0.1 μ s duration (the inverse of the 10 MHz bandwidth). Further increase in the signal-to-noise ratio (S/N) of the processed radargrams is achieved through synthetic aperture radar (SAR) processing, which also narrows the along-track

footprint to a few hundred meters. Radar vertical and horizontal resolutions are, respectively, 15 m (free-space) and 3–6 km (cross-track) by 0.3–0.6 km (along-track) (Seu et al., 2007b; Biccari et al., 2002; Croci et al., 2011). The spacecraft reference frame in flight has the *X*-axis oriented toward the direction of flight and parallel to SHARAD's antenna and the attachment points of the solar arrays. The *Z*-axis points to the planet (nadir-looking direction). The *Y*-axis is in the cross-track direction and completes the frame. During nominal science operations, the spacecraft can be rolled around the *X*-axis to the left or right up to 28° from nadir, to satisfy targeting requirements of the various science instruments. SHARAD is not a directional instrument and its observations always produce along-track radargrams, with the signal largely returned from nadir. Such rolls change the orientation of the antenna radiation-pattern with respect to the surface, and rolls to positive angles produce an increase of gain and thus have been used routinely to improve the quality of the observations whereas negative roll angles have a very detrimental effect on the gain. From analysis performed during SHARAD development (Croci et al., 2007), it is also

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Table 1Thicknesses and electromagnetic parameters, at $f = 20$ MHz, for the layers used to model solar arrays.

Layer	Thickness (mm)	ϵ_r	σ (S/m)
Aluminum honeycomb (3003 alloy, 28.8 kg/m ³)	19.050	–	–
Kapton	0.051	3.3	0
GaAs solar cells	0.190	13.0	1.8E+4
Glass	0.152	4.5	0

known that the antenna pattern is affected by the position of the two very large solar arrays (SA) and a special configuration designated “SHARAD SA Park” is used when the spacecraft is in eclipsed conditions to maximize SHARAD performances during nightside segments of the orbit. The SHARAD SA Park position was determined on the ground using test range data acquired during instrument development and then validated in flight after the antenna deployment (the full definition will be given in Section 3). The potential impact of the interaction between antennas and solar panels at megahertz frequencies has been an important consideration for several space missions, including MRO and the JUICE mission to Jupiter. Previous studies have shown that this interaction can lead to changes in the performance of wireless communication systems, which sometimes have been mitigated by changes in solar panel design, and in particular in their grounding strategy, to minimize such interaction (Bruzzone et al., 2013; Hahnel et al., 2018a,b, 2022). Unlike other spaceborne sounder antennas at megahertz frequencies, the Mars Advanced Radar for Subsurface and Ionosphere Sounding (MARSIS) and the Apollo Lunar Sounder Experiment (ALSE) were deployed on spacecraft bodies and with solar arrays that are significantly smaller than the sounder antenna, thereby mitigating potential radiation pattern impairments (Porcello et al., 1974; Macher et al., 2006-01). An analysis of spacecraft and solar array interaction with the sounder’s antenna was performed also for the MARSIS instrument, but without considering solar array movements (Jordan et al., 2009). As a preference, SHARAD is operated during the nightside segment of orbits to reduce also the effect of Mars’ ionosphere, but dayside operations are also performed routinely for targeting convenience and for assessing ionospheric effects. MRO orbits Mars with a sun-synchronous orbit set at 15:00 local mean solar time (at ascending equatorial crossing). The orbit altitude varies between 252 km and 319 km with the periaresis in the area of the south polar cap. The reference attitudes of the spacecraft is shown in Fig. 1. SHARAD and its antenna are mounted on the +Y edge of the space-facing -Z deck which is usually permanently in shadow because of the LVLH (Local Vertical Local Horizontal) orientation of the spacecraft during nominal science operations. Other science instruments are mounted on the +Z deck that faces the surface of the planet (nadir direction). The peculiar position of SHARAD, clearly not ideal for sounding of the Martian surface, was chosen due to important spacecraft priorities and constraints also linked to the 10 m length of the deployed antenna.

2. HF Antenna model

As predicted during mission design, there is significant electromagnetic interaction between the SHARAD 10-m, center-fed, dipole antenna and the large Solar Arrays (SA) and High-Gain Antenna (HGA) of MRO. The effects of such an interaction were partially characterized by ground simulations on a scaled mock-up of the satellite. During the early phase of the mission, the effects of the SA and HGA were evaluated by measuring the nadir-direction echo at different spacecraft attitudes (Crocchi et al., 2007).

More recently, Campbell et al. (2021a) analyzed data for the entire mission and proposed an empirical model of SA and HGA effects on radar performances. Through the systematic comparison of neighboring SHARAD tracks that were collected under varying SA and HGA orientations a 4 dB spread in surface power was observed. The evaluation of spatially approximate data was undertaken to mitigate the influence of surface properties on the power values recoded by SHARAD. Campbell

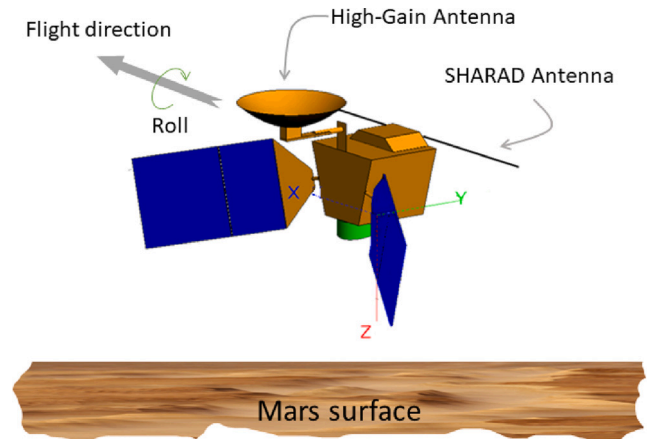


Fig. 1. Mars Reconnaissance Orbiter (MRO) nominal orientation. 10-m dipole antenna and all the other features implemented in the MoM model are shown. Solar arrays are shown in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

et al. (2021a) analysis also confirmed antenna gain increases associated with positive roll maneuvers, the maximum allowed being $\theta = 28^\circ$, and established that such improvements are also modulated by orientations of SA and HGA. However, science data estimates and earlier modeling suggested that greater roll angles would produce additional benefits. This possibility prompted the development of a new, detailed, electromagnetic model able to predict the SHARAD antenna pattern for any MRO attitude and for any SA and HGA orientation. In response to the request from the SHARAD science community, led by Prof. Roberto Seu (Uniroma1/DIET) and Dr. Nathaniel Putzig (PSI Colorado), the first two authors of this communication developed a Method-of-Moments (MoM) model of the SHARAD dipole antenna.

Most parts of the spacecraft were modeled as a Perfect Electrical Conductor (PEC), with the exception of the solar arrays. The antenna is modeled as a single conductor having a radius of 3.9 mm. This value is the radius equivalent to two AWG22 conductors placed with a spacing of 48 mm (Balanis, 2015). The tip-to-tip length is 10.1 m.

Each SA panel is modeled as a multi-layer structure: an aluminum honeycomb core (wall thickness of 50 μ m, cell diameter of 20 mm) sandwiched between composite facesheets, a Kapton sheet, and a solar cell layer, as described in Dec et al. (2007), which however does not provide the thickness of layers. Therefore, the detailed description of the stack-up is taken from Dec and Amundsen (2003) and consists of aluminum honeycomb, Kapton, gallium arsenide (GaAs) solar cells, and glass layers. Their parameters are listed in Table 1 and are given for $f = 20$ MHz. Composite facesheets and adhesive layers are not included in our SA model, being very thin. The response of the SA stack-up to a generic impinging electromagnetic wave was first modeled using Ansys HFSS (HFSS, 2023), where a unit-cell of hexagonal shape, subject to an impinging wave, is analyzed by the Finite Element Method (FEM). Results for both TE and TM polarizations are shown in Fig. 2, together with a simplified model that assumes the honeycomb layer being a bulky homogeneous material. Electrical conductivity for this honeycomb-equivalent material has been set in order to obtain a good fitting with FEM results for the actual honeycomb. A homogeneous

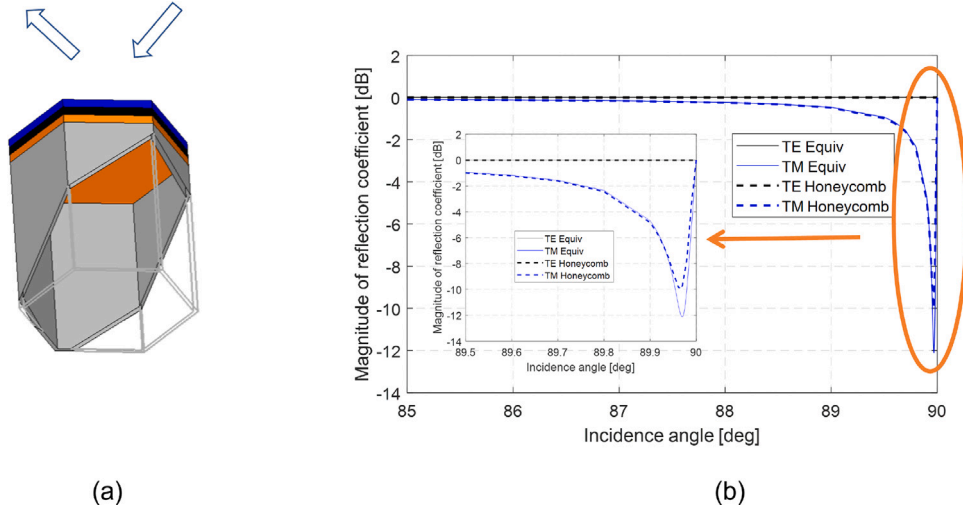


Fig. 2. Solar Array as a multi-layer periodic structure, modeling an aluminum honeycomb core covered by Kapton, gallium arsenide and glass. (a) unit-cell, for which the actual layer thicknesses are provided in Table 1; (b) magnitude of reflection coefficient, in dB, for both TE and TM polarization, as obtained by the FEM model of the unit-cell and by a simplified transmission-Line model. The latter models the honeycomb layer as a homogeneous material with an electrical conductivity that gives the same pseudo-Brewster angle as the FEM model does. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

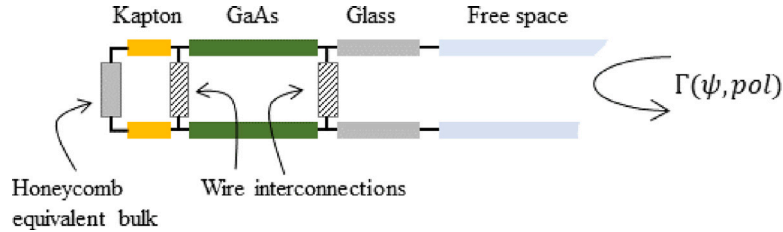


Fig. 3. Transmission-line model of SA stack-up. It allows computation of the reflection coefficient Γ for any incidence angle ψ and for $pol = \{TE, TM\}$. The impedances due to wire interconnections are not modeled here, due to the very low impedance seen at any transmission-line interface.

honeycomb-equivalent medium is particularly useful for both the implementation of a Transmission-Line (TL) equivalent circuit (Ramo et al., 1994) and the overall MRO model.

The TL equivalent circuit models the honeycomb-equivalent bulk medium as a lumped impedance $Z_t = (1 + j) \sqrt{\frac{2\pi f \mu}{2\sigma_{eq}}}$, where $\sigma_{eq} = 10^3$ S/m. At 20 MHz, $Z_t = 0.28(1 + j) \Omega$. The TL model could also include the presence of arbitrary wire connections at the interface between adjacent layers, as depicted in Fig. 3. The effects of these wire connections will be, however, neglected considering that they are shunt connected with the very low impedance of the remaining parts of the model of Fig. 3. The TL model returns the reflection coefficient Γ for any plane wave impinging on the SA. Gallium Arsenide conductivity has been set to the value that gives the worst case (i.e. the lowest magnitude of Γ), following the same approach as in Yekan and Baktur (2021).

The HiRISE (High Resolution Imaging Science Experiment) camera baffle was modeled as a PEC surface, due to the metallic thermal blanket that covers most of it.

Even considering losses due to the solar arrays, modeled according to the approach described above, the efficiency of the radiating system is very close to 100%, ranging, when in SHARAD SA Park position, from 99.6% to 99.8%, corresponding to -0.017 dB and -0.009 dB, respectively. For this reasons we assume here that directivity and antenna gain are the same.

The adopted numerical model is based on MoM implementation of the Electric Field Integral Equation, which is solved in the frequency domain using the ALTAIR-FEKO suite (FEKO, 2023). The overall MRO structure is modeled adopting a “fine” mesh (about 6000 triangles, having sides no larger than $\lambda/16$ and 50 segments no longer than $\lambda/25$, where λ is the wavelength at 25 MHz), achieving the best accuracy.

3. Numerical results

One of the key factors in radar S/N is the antenna gain in the nadir direction for a given MRO attitude. In this study, like in nominal MRO operations, only a variation of the roll angle (around the X-axis) is considered, maintaining the pitch and yaw rotations at 0° . At 0° roll angle, the nadir direction coincides with the +Z direction of Fig. 1. When the roll angle is set to an angle θ_0 , the nadir direction becomes the $(\theta_0, 90^\circ)$ direction of the polar reference frame associated to the (X, Y, Z) one.

Another key factor in radar S/N is the position of MRO appendages, namely the two solar arrays (SA Plus X, or SA_{PX} , and SA Minus X, or SA_{MX}) and the communications High-Gain Antenna (or HGA). Each appendage has two rotation gimbals, an Inner Gimbal (IG) and an Outer Gimbal (OG). When MRO is not eclipsed by Mars, the solar arrays track the Sun (to maintain battery charge), and when Earth is not occulted by Mars, the HGA tracks the Earth (to ensure communications). At orbital night and when Earth is occulted, the solar arrays are usually “parked” in a position beneficial to SHARAD that is referred to as the SHARAD SA Park position, and the HGA is put at a rest orientation of $HGA-IG = 0^\circ$ and $HGA-OG = 90^\circ$. The two conditions happen mostly in the same portion of the orbit but with different beginning and ending times.

In total there are 6 gimbals (see Fig. 4) that act as parameters affecting SHARAD’s antenna pattern, but this study confirmed that the HGA position has no significant effect, as shown in Fig. 5, where the 20 MHz radiation patterns are compared for two different HGA positions, when SA are in SHARAD Park position. In addition, the $SA_{PX}-OG$ is experiencing wear issues and therefore it is now kept in the 0° position. This leaves only three parameters to deal with in modeling the effect on the radiation pattern: $SA_{PX}-IG$, $SA_{MX}-OG$ and $SA_{MX}-IG$. This study

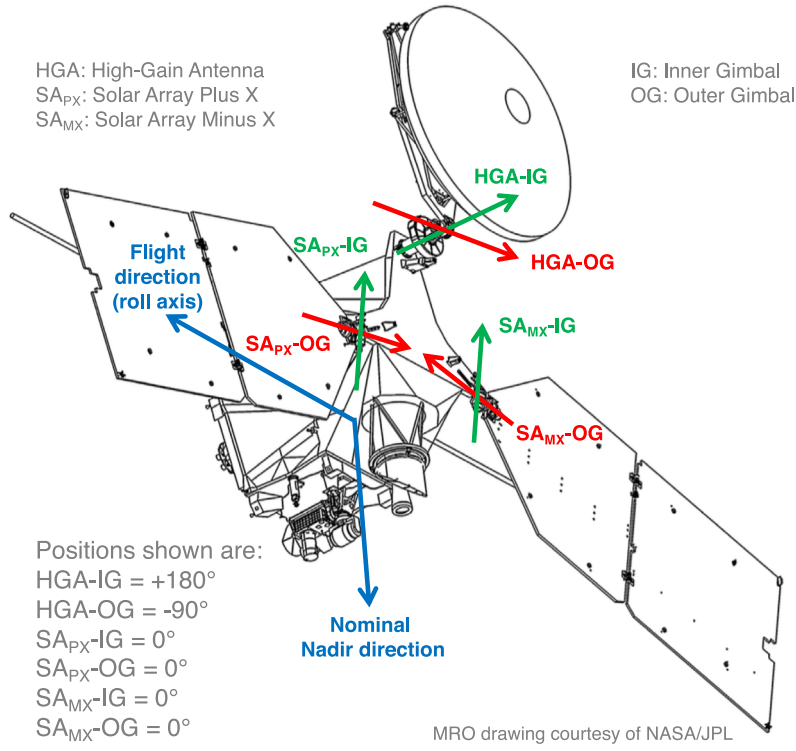


Fig. 4. MRO's High Gain Antenna and Solar Array orientations with gimbal rotation axes.

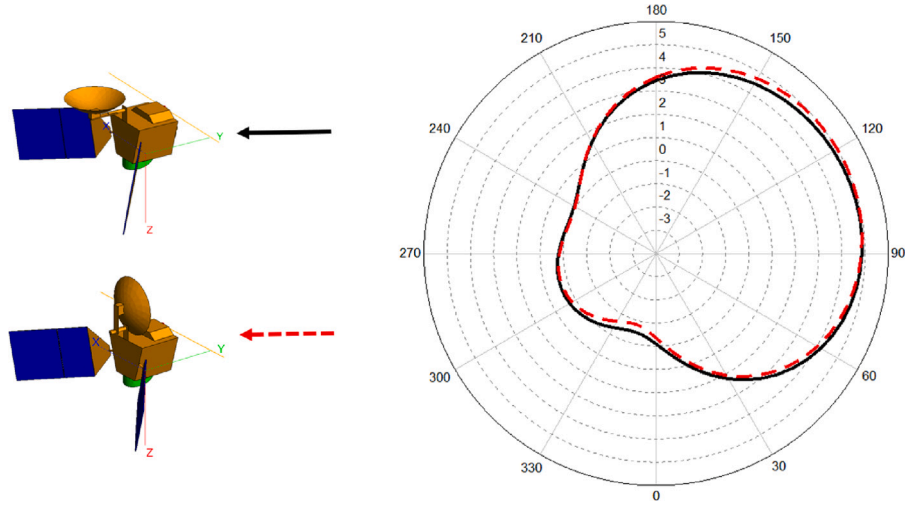


Fig. 5. Polar plot of Antenna Gain (dBi, Y-Z plane cut) vs. roll angle at $f = 20$ MHz, for two different angles of HGA-OG. No significant effects appear.

considered in particular the configuration with all gimbals at 0° and the SHARAD SA Park Position used at night and defined as follows:

$$\begin{aligned} SA_{PX}-IG &= +28.36^\circ \\ SA_{PX}-OG &= 0^\circ \text{ (it used to be } -9.96^\circ) \\ SA_{MX}-IG &= -37.98^\circ \\ SA_{MX}-OG &= 9.96^\circ \end{aligned}$$

SA-OG gimbals have the effect of revolving the SA around their longitudinal axes, without significantly modifying the mutual distances between SHARAD antenna and the SA themselves. So, changing SA-OG gimbal angles has a minor effect on radiation pattern and gain. Thus, in the following discussion, both SA-OG values are set to 0° . Therefore, the different SHARAD configurations discussed below are based only on

the values of the X-axis inner gimbal angles ($SA_{PX}-IG$ and $SA_{MX}-IG$) of the two solar array panels (see Fig. 4).

An example of the modeled antenna pattern, for a specific setup of SA defined by the two inner gimbal angles $SA_{PX}-IG$ and $SA_{MX}-IG$, is shown in Fig. 6(a). Radar Antenna directivity at nadir enters twice in the S/N expression (because of the nature of radar operations). So we call “round-trip radar gain” a value of twice the antenna directivity at nadir. It is thus possible to draw the plots of Fig. 7, where the increments of round-trip radar gain (RTGI) with respect to a given reference configuration are shown as a function of roll angle and of other relevant configuration parameters (i.e. $SA_{PX}-IG$ and $SA_{MX}-IG$). The reference configuration is all zero gimbal angles and zero roll angle. The specific roll angle values used for the different plots have been chosen to show the difference between: no roll (S/C nadir looking), minimum useful roll angle ($+20^\circ$) during nominal operations,

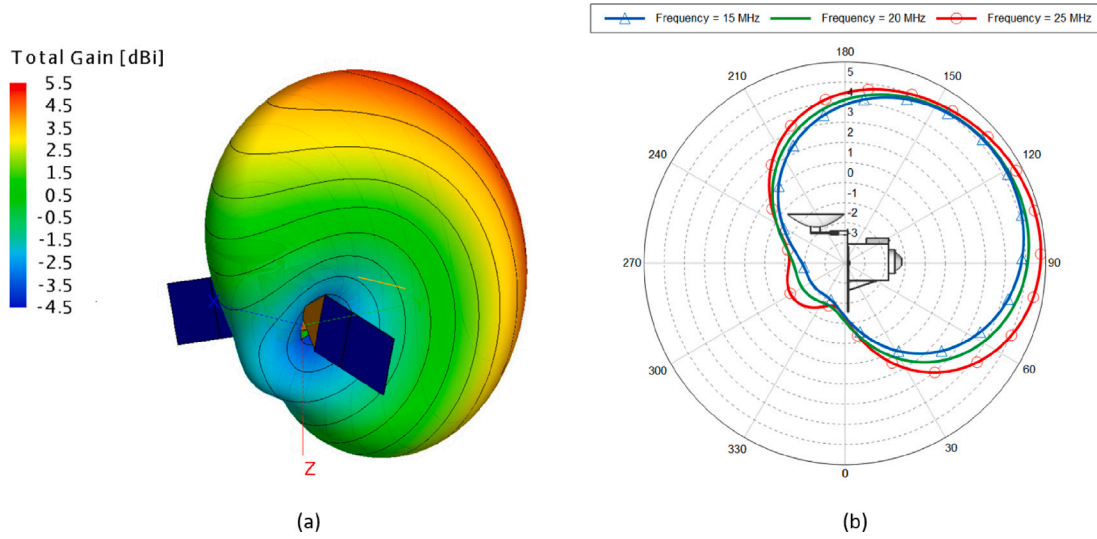


Fig. 6. (a): Radiation pattern at $f = 20$ MHz for all gimbal angles set to zero. Maximum antenna gain of 5.3 dBi is achieved at $\theta = 120^\circ$, while at $\theta = 0^\circ$ the antenna gain is -1.1 dBi. (b): polar plot of Antenna Gain (dBi) vs. roll angle θ from +Z axes, for directions belonging to Y-Z plane and for $f = \{15, 20, 25\}$ MHz. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

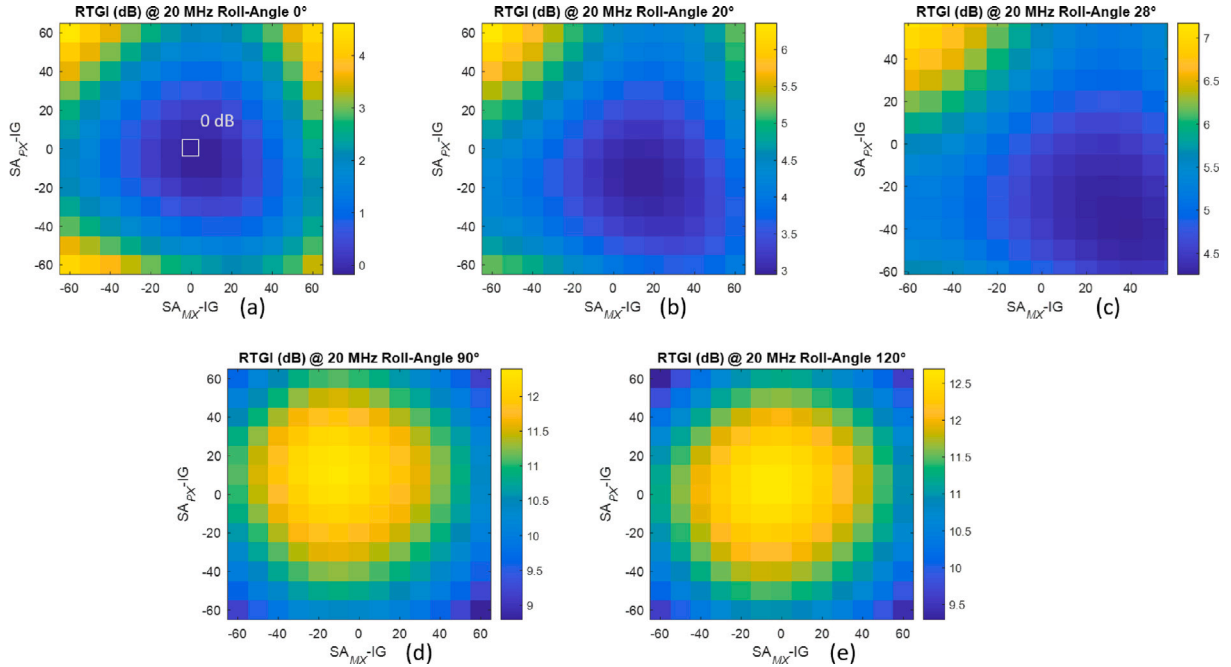


Fig. 7. Round-Trip Radar Gain increments (RTGI in dB) for some allowed couples of SA_{MX-IG} and SA_{PX-IG} internal gimbal angles. Subplots (a) to (e) refer to roll angles of $\{0^\circ, 20^\circ, 28^\circ, 90^\circ, 120^\circ\}$, respectively. 0 dB reference is indicated in (a). Best results are obtained with $\theta_0 = 120^\circ$ roll angle. Please, note the different value ranges for each panel.

maximum roll angle ($+28^\circ$) during nominal operations, best values for increased gain operations ($+90^\circ$ and $+120^\circ$).

The largest RTGI is observed for a roll angle $\theta_0 = 120^\circ$ and $SA_{PX-IG} = 0^\circ$, $SA_{MX-IG} = 0^\circ$. A first validation of the present electromagnetic model has been performed by comparing the empirical based model of Campbell et al. (2021a) with the predictions presented in this paper (Fig. 8). The two plots in Fig. 8 show good agreement in terms of the broad range in gain that is to be expected (though note the electromagnetic model predicts a gain improvement 0.5 dB higher than the empirical model) as well as the gain variations with respect to SA_{IG} angle. The electromagnetic model predictions do however, show greater detail in gain variation relative to the empirical based model as a subtle asymmetrical pattern in gain variation is observed in Fig. 8(a), with the greatest improvement associated with a SA_{MX-IG} of

-60° and a SA_{PX-IG} of 60° . The higher fidelity of the electromagnetic model predictions is expected given that the derivation of the empirical model had to contend with additional influences beyond SA and HGA orientations, such as surface roughness and reflectivity and a stochastic space environment.

Additional plots to show the gain versus the roll angle in a given configuration have been also generated considering also the behavior not only at SHARAD's central frequency of 20 MHz, but also at the edges of its operative bandwidth, i.e. 15 MHz and 25 MHz. In Fig. 6(b), the clear advantage of a roll at 120° for the configuration with all gimbals set to 0° is evident. The model predicts that this gimbal configuration is the optimal one for high-angle roll conditions, as opposed to the SHARAD SA Park Position used at night in nominal conditions, which is preferred at lower roll angles. To fully assess the optimal

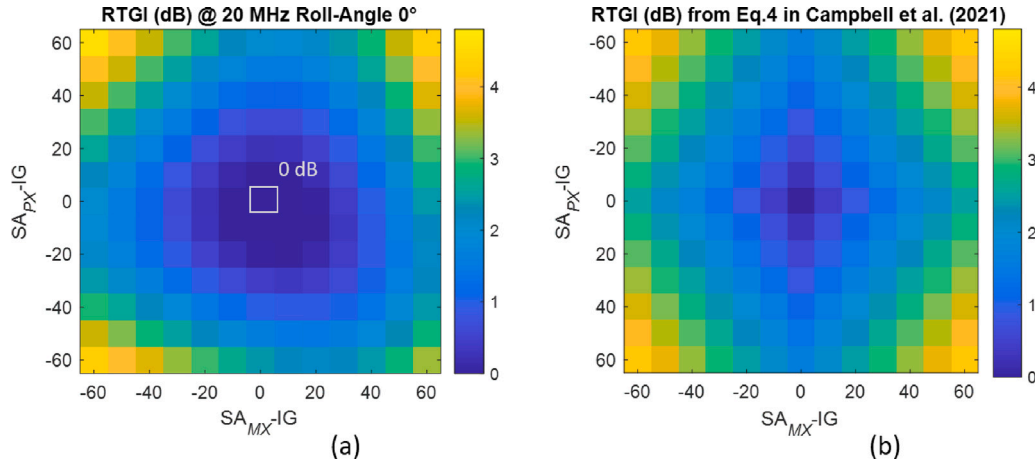


Fig. 8. Comparison of Round-Trip Radar Gain increment (RTGI in dB) predicted by the present model (a) and from the empirical observations (b) as from Eq. 4 in Campbell et al. (2021a). Gain at ($SA_{PX-IG} = 0^\circ$, $SA_{MX-IG} = 0^\circ$) is considered as 0 dB reference.

roll angle we considered not only the behaviour at SHARAD's central frequency of 20 MHz, but also at the edges of its operative bandwidth, i.e. 15 MHz and 25 MHz. Fig. 6(b) demonstrates how a roll at 120° (for the configuration with all gimbals set to 0°) provides an ideal compromise between achieving maximum gain while also ensuring a balanced response across all transmitted frequencies. SHARAD uses a Linear Frequency Modulated ("chirp") pulse with a bandwidth of 10 MHz to achieve a theoretical resolution (in free space) of 15 m. The actual resolution, after the heavy Hann weighting applied to reduce sidelobes, is 25 m. At 120° roll angle, the roll-off in frequency is in the order of 1 dB (one-way) as shown in Fig. 6(b). The resulting impact on the system impulse response is a slight degradation in the range resolution, negligible with respect to the one due to weighting. Critically, the modeling also reveals the dynamic relationship between gain variation as a function of both roll angle and SA configuration. The empirical-based analysis of Campbell et al. (2021a) found SA orientation to be critical to the gain improvement during rolled maneuvers up to the previous maximum permitted roll of 28° . The results presented in Fig. 7 reveal that the SA orientation needs to also be seriously considered for more extreme rolls. Compare the nadir situation presented in Fig. 7(b) with that of the 120° roll in Fig. 7(e). The optimal gain for both roll settings are essentially inverse to one another, with 0° gimbal preferred for the latter, high roll scenario. Failure to adequately account for SA gimbal angles could result in a different of about 3.5 dB in the potential gain achieved, meaning that mission operations preparations for high roll maneuvers need to carefully account for SA orientation as part of the planning procedure process.

4. In flight validation

To assess the results of the electromagnetic model of the SHARAD antenna, and to confirm the expected science benefits from an observation performed at the predicted optimal orientation of the antenna with respect to the Martian surface, the instrument operations team proposed a Very Large Roll (VLR) Test to the MRO managers.

Due to the compromised positioning of the SHARAD antenna on the upper corner of the spacecraft bus, the instrument obtains relatively low S/N at 0° roll angle. MRO mission management presently allows rolls up to 28° to enable a substantial improvement (up to 6 dB, round-trip) in SHARAD data (see Fig. 9), and the SHARAD team regularly seeks to obtain such rolled observations when compatible with other science operations. With the prospect of obtaining significant additional S/N improvement at much larger roll angles, the SHARAD team proposed a test involving a 120° roll angle. Such a roll angle required a special operations sequence to be inserted in the standard weekly

operations sequence, with particular attention to different spacecraft constraints: power, thermal, star trackers field-of-view, and more.

After a long phase of proposed target selections, feasibility studies of the off-nominal spacecraft sequences, simulations on the orbiter test-bed at Lockheed Martin, and in depth analysis of all risks associated, the test was approved and executed during the first week of May, 2023.

For the VLR Test observation (78583-01), the selected ground track crossed the equator from 13.45° N to 35.8° S (see Fig. 10). This orbit was chosen as it covers a high priority science target that has been the subject of previous SHARAD studies and thus provides an effective site with which to assess the performance of the VLR. An additional comparison observation, for a ground track with the same latitude extent as the test one, but shifted one degree to the right, was acquired two weeks before. Other previous observations covered the area of interest, and were selected for comparison.

To effectively assess the gain improvements facilitated by the test required a region with well established subsurface structure that was also in eclipse at the time of data collection, so as to avoid ionosphere induced effects on the S/N (operational considerations regarding the impact to science observations during the daytime portion of the orbit, carried out by the other instruments, also required the test to be conducted at night). Given that the timing of the test corresponds to northern summer on Mars where the beginning of the eclipse period for MRO's orbit extends to lower northern latitudes, the team chose to target the equatorial Medusae Fossae Formation (MFF). The MFF is a kilometer-thick accumulation of ancient (at least Hesperian in age) layered deposits of unknown origin. Previous radar sounder studies using MARSIS and SHARAD data have detected basal reflectors below the deposits and corresponding analyses show that they have low bulk permittivity and low loss, suggestive of the presence of ice at depth (Watters et al., 2007; Carter et al., 2009; Campbell et al., 2021b). Other studies have argued against an ice-rich origin for the MFF and instead suggest they are a product of volcanic activity (Bradley, 2002; Hynek, 2003; Kerber et al., 2011; Zimbelman and Griffin, 2010; Mouginiis-Mark and Zimbelman, 2020; Ojha and Mittelholz, 2023). Currently, SHARAD penetration of these materials is limited, but the improved S/N from a VLR may allow probing of these deposits to greater depth and perhaps reveal internal structure. Consequently, an observation at a high roll angle over the MFF not only satisfies the gain test requirements but also provides data of high scientific value to help reconcile the competing hypotheses of MFF genesis.

Some of the deepest reflectors detected by SHARAD outside of the polar regions occur below Eumenides Dorsum (Fig. 10) (Campbell et al., 2021b), where a basal reflector of 10 μ s delay from the surface return is present in previous observations. The particular orbit chosen for the test closely repeats portions of the ground tracks of two earlier

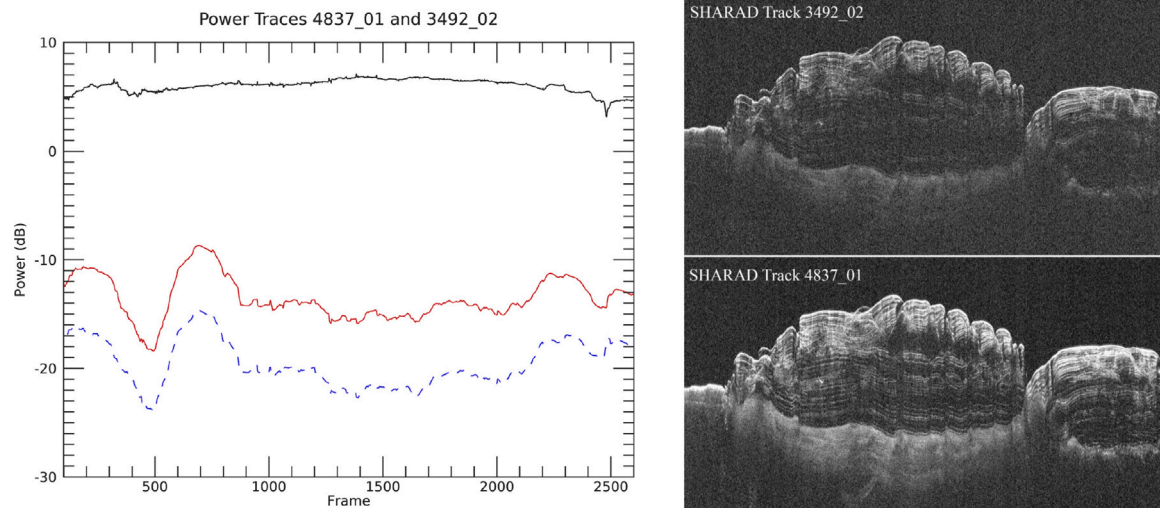


Fig. 9. Left panel shows surface echo power traces for two SHARAD observations of the north polar layered deposits. Red solid trace collected with 22° – 26° roll angle, blue dashed trace with no roll. Black trace at top is the difference in power between the observations, which shows a mean echo enhancement of 6 dB. Right panels show the corresponding radargrams, where upper radargram is the one obtained with no roll. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Source: Reprinted from Campbell et al. (2021a).

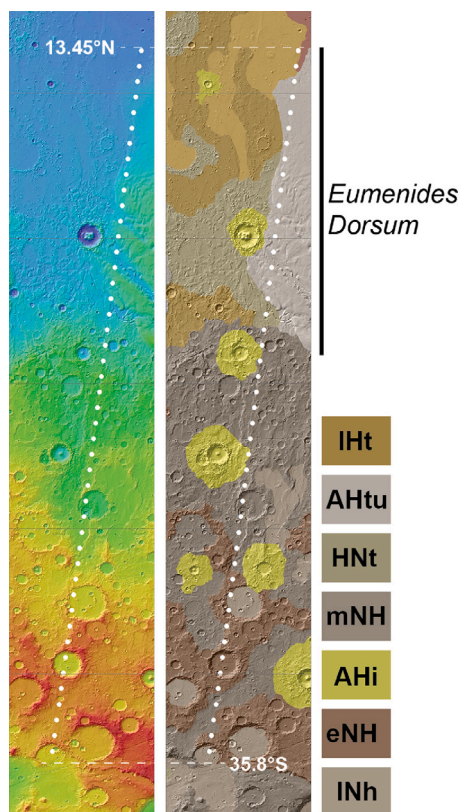


Fig. 10. Orbit selected for the VLR Test, highlighting the start and stop limits of the radar acquisition. The flight path is descending (night portion of the orbit) from North to South. Two different backgrounds are presented: one over MOLA altimetry (left) and one over a geological map from Tanaka et al. (2014) (right). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

observations obtained at 0° roll angle (SHARAD observations 57257-02 and 20283-01). As the basal reflector extends laterally for 300 km and is highly dipping, it provides an excellent opportunity to examine

the S/N improvement from both surface and subsurface reflections of varying depth. Given the restriction precluding more northerly targets, the test was extended into the southern mid-latitudes over highland terrain. This region hosts a variety of terrain types (see again Fig. 10) that exhibit a range of surface textures and roughness that will provide further test of the surface response.

The VLR Test observation was successfully acquired on 2023-05-02 between 11:22:00 and 11:38:00 UTC. After the data were made available on ground, they have been decoded and processed to produce a first estimate of the results.

A qualitative visual comparison of the radargram from the VLR Test with those of the previously acquired observations at 0° roll angle shows that the VLR radargram appears much brighter than the prior ones. In general, the nadir surface return, apparent subsurface returns, and clutter signals all appear to have significantly more power in the VLR radargram, and the occurrences of clutter extend to significantly greater delay times. In regards to the Medusa Fossae area (see Fig. 11), one may follow the basal return more continuously under the northernmost lobe of Eumenides Dorsum, and fainter subsurface returns are also now evident below the outlying low mound to the north. These results demonstrate the clear advantage of VLR observations in illuminating features with higher S/N and at greater depth than is achievable at lower roll angles.

The surface power returns were extracted in portions of the radargrams that overlapped regardless of whether the VLR observation had reached its optimal viewing conditions since even in sub-optimal viewing the VLR observation would have superior SNR.

An initial quantitative assessment, (see Fig. 12) has been performed by comparing the surface power return of the VLR observation and those from coincident SHARAD observations 57257-02 and 20283-01. The two previous observations were both nadir observations collected in 2018 and 2010, respectively. 57257-02 starts further to the north and overlaps with the beginning of the very large roll observation while 20283-01 begins roughly in the middle and extends well south of the very large roll observation. To extract the surface power returns, the nadir arrival times estimated through clutter simulations were used. These arrival times are typically a few samples off from the true signal arrival due to the idealized nature of the clutter simulations. Therefore, we employed a quick correction to the nadir returns by defining a 6-sample window around the estimated nadir arrival time and extracted the maximum power within that window. To estimate the noise level in

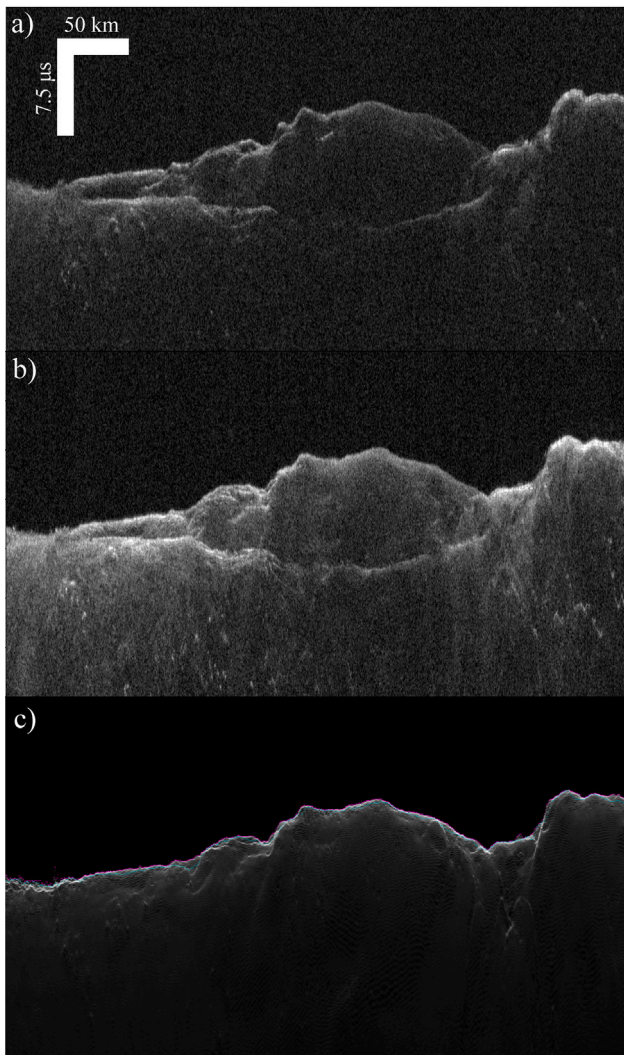


Fig. 11. Comparisons of (a) radargram for a portion of 0° roll angle SHARAD observation 57257-02, (b) radargram for a portion of VLR SHARAD observation 78583-01, (c) and a portion of the clutter simulation for SHARAD observation 78583-01 over Eumenides Dorsum of the Medusa Fossae formation. The clutter simulation helps distinguish true subsurface returns from surface clutter that, with the additional gain resultant from the 120° roll, becomes more prominent. The estimated nadir returns (cyan) and first returns (magenta) are also shown on the clutter simulation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

each data set, the same nadir arrival times were shifted by 500 samples up-range and the noise level was extracted. These extracted power values were then binned using 1-dB bins that spanned each data set.

The total gain was compared with the predictions performed before the observation by using the model. The predictions considered also the changing attitude of the solar arrays during the initial minutes of the observation, as shown in Table 2. This movement, which brought the SA to the ideal attitude (for a VLR) of 0° on all gimbals, has been necessary to optimize both the extent of the observation northward and the need to keep the battery under charge until the very start of the eclipse portion of the orbit.

These initial numerical assessment provided a very good result when compared with the model, when considering the stable configuration. For a more precise comparison during SA movement a more in depth analysis will be required. As the values reported in Fig. 12 show, a total gain increment of around 10 dB has been demonstrated, in line with model predictions, with only a minor increase in the total

noise distribution. The observed 0.9 dB increase in the average noise level is well within the variation in all SHARAD data, which can vary by up to 5 dB (Campbell et al., 2021a). This reflects also an increase of around 9 dB in the total S/N, an extremely important result for future science observations. To further assess the performance of the VLR observation, scientific analysis of the subsurface return of the MFF should be conducted, the results of which should be compared to previous results to ascertain the ability of VLR observations to provide further constraints on existing scientific questions.

In summary the VLR test has demonstrated that the predicted 10 dB gain enhancement for a roll of 120° with preferential positioning of the solar arrays (Fig. 6) was achieved (Fig. 12). This near order of magnitude increase in gain resulted in significantly brighter surface (both nadir and off-nadir, i.e. clutter) and subsurface returns. As anticipated, the VLR also facilitated greater penetration into the subsurface enabling deeper basal interfaces to be recorded within the MFF deposits relative to equivalent 0° roll collects (Fig. 11).

5. Future benefits for SHARAD science

The VLR test showed the expected improvement in performances and S/N, and the SHARAD Team is already in discussion with the MRO Project to consider occasional use of such large rolls to obtain additional high S/N observations over targets of high scientific interest. Some of the complexities that arose during the planning of the first test will be mitigated in future observation by concentrating the areas of interest to smaller ranges of latitudes. One target of high scientific interest is the south polar region's Planum Australe, where substantial attenuation of SHARAD signals occurs, including above the location of strong basal returns obtained by MARSIS that has been interpreted as associated with liquid water in several locations (Orosei et al., 2018; Lauro et al., 2020). It is anticipated that the improved signal strength may allow SHARAD to obtain reflections to this same depth and thereby be able to contribute to the assessment of these features at a different radar wavelength. SHARAD signals are also often attenuated in other areas of the south polar region prior to reaching the base of the icy layered deposits (Whitten and Campbell, 2018), and VLR observations may enable detection of that base as well as shallower interfaces above it that have yet to be examined. While SHARAD has had greater success in illuminating deeper structures in the north polar region, much of the main lobe of Planum Boreum contains a highly attenuating set of basal units that only MARSIS has thus far been able to penetrate consistently. If a VLR observation over this region is able to obtain a reflection from the base of the basal units, new information about the nature of these deposits, perhaps including details of internal layering and dielectric properties that inform ice purity, may be forthcoming. For both polar regions, the improved S/N afforded by VLR observations are likely to lead to a better understanding of signal attenuation through the stacks of previously illuminated layers and thus providing new information about their compositions. Beyond the polar regions, there are several mid-latitude sites, including those containing debris-covered glaciers (Petersen et al., 2018), ground ices (Bramson et al., 2015; Sturman et al., 2016), and layers of volcanics and sediments (Campbell et al., 2008; Morgan et al., 2015), that represent promising targets for VLR acquisitions. In many of these areas, the subsurface reflections obtained at lower roll angles are relatively weak and often limited to a single interface. We expect that VLR observing will yield a significant improvement in the S/N for the known interfaces and perhaps illuminate additional, deeper interfaces in these terrains. In some instances, dipping subsurfaces interfaces in current observations at lower roll angles are no longer found beyond a certain depth (e.g., Campbell et al., 2008), and some extension of detectability to greater depths is anticipated.

We anticipate that specific science investigations will therefore benefit greatly from VLR observations, but the accuracy of the newly computed antenna model extends also to other aspects. In particular it

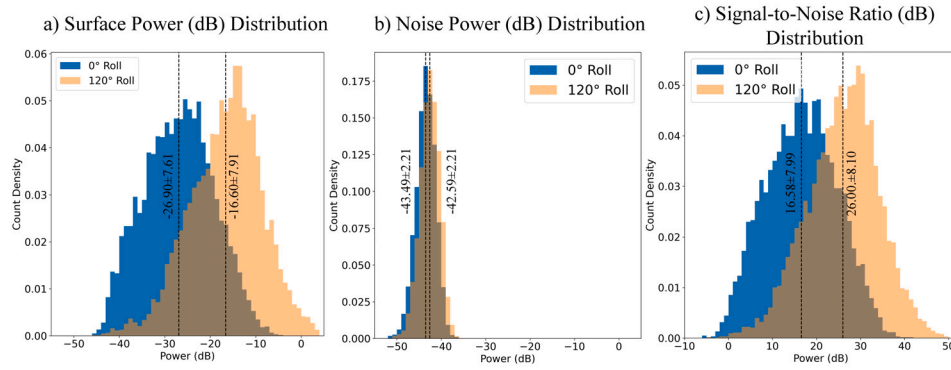


Fig. 12. Histograms showing the distribution of the (a) surface power returns, (b) noise level, and (c) signal-to-noise ratio extracted from coincident observations 57257-02 and 20283-01 (blue) and 78583-01 (VLR, orange). All histograms were calculated using 1 dB bin widths spanning each data set and the means (dashed black lines) and standard deviations of each are displayed.

Table 2

Predicted solar array gimbal angles, in degrees, and the corresponding antenna gains, during the VLR Test.

Time (UTC)	Predicted SA angles				One-way antenna gain		
	SA _{PX} -IG	SA _{PX} -OG	SA _{MX} -IG	SA _{MX} -OG	15 MHz Gain (dBi)	20 MHz Gain (dBi)	25 MHz Gain (dBi)
May 02, 2023							
11:08:01	28.42	0.00	64.67	-9.55	3.8	4.2	4.9
11:20:01	25.47	0.00	61.85	-7.24	3.9	4.3	5.0
11:21:01	13.66	0.00	45.90	-3.60	4.4	4.8	5.3
11:22:01	6.80	0.00	23.05	-1.80	4.9	5.1	5.4
11:23:01	3.39	0.00	11.38	-0.89	5.1	5.3	5.5
11:24:01	1.69	0.00	5.67	-0.45	5.2	5.3	5.6
11:25:01	0.84	0.00	2.82	-0.22	5.2	5.3	5.6
11:26:01	0.42	0.00	1.41	-0.11	5.2	5.3	5.6

is worth mentioning that the huge existing data set can now be revisited with a better knowledge of the antenna gain during the observation (in many cases the spacecraft configuration changes during an observation, both in roll and in the positioning of the SA and HGA appendages). By tagging collected SHARAD data with the anticipated gain of the observation (relative to the null value at 0° roll), will provide users with a better appreciation of the relative data quality for a given scientific endeavor. For example, in the case of investigating highly attenuation terrain such as volcanics, one would be cautious to rule out the presence of deep reflectors within the region if all of the data coverage was acquired with a low gain configuration. Likewise, the derivation of global products, such as surface reflectivity maps (e.g. [Campbell et al., 2021a](#)) rely on the effective normalization of the influence of observational parameters, which could be greatly improved by the computed antenna model presented here. In addition, SHARAD targeting and planning activities, managed weekly, will benefit from prediction of the gain available during the execution of future observations to better prioritize the selection of suitable science targets.

6. Conclusions

The electromagnetic model of Mars Reconnaissance Orbiter HF Radar antenna and its interaction with the spacecraft, developed under the study presented in this communication, is an important step for improving SHARAD science. The validation of the model has already proved successful, but a more in-depth analysis remains to be carried out. Practical uses of the model will extend also to nominal targeting operations, the analysis of the data already available and other possible applications. Planning VLR observations requires a dedicated spacecraft commands sequence and the careful assessment of thermal constraints (if roll started before eclipsed conditions) and power margins, not to risk discharging spacecraft batteries too deeply. These constraints are more critical at higher latitudes. Ideally, VLR observations will be performed in eclipsed condition with at least 10 min of margin from the beginning and ending of the eclipsed portion of the ground

track. Operational difficulties in repeating this kind of VLR observations can be partially overcome now that the model has been verified at a high roll angle, for example by limiting observations only to high-priority investigation areas and keeping acquisition away from terminator crossings. In general, the capability to perform observations at a VLR on selected targets will open new avenues of research and it is expected to produce new scientific discoveries that may lead not only to a better comprehension of Mars geophysical processes, but also to the selection of human landing sites.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The radargrams are available on the PDS or through CO-SHARPS. The antenna modeling data could be made available upon request.

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